



INFORMATION SYSTEMS | EDUCATIONAL VIDEO PROJECT

Problem Solving in Education

A step-by-step guide to creating your educational video



Tool



5 Stages



Real Problem



Barriers

WHAT IS THIS PROJECT?



THE SCENARIO

You have been approached by the Information System department manager to help create an educational video. You must record a video explaining and demonstrating how a problem can be solved in an educational setting (such as a school, classroom, or college).



Video Length

5 – 7 minutes



Format

MP4 or similar



You Must Appear

On camera, speaking clearly in English

YOUR VIDEO HAS 4 PARTS

1

Problem-Solving Tool

25 marks

Name a tool (e.g. flowchart), explain how it works, give a school example.

2

Stages of Problem Solving

20 marks

Explain all 5 stages clearly with a short description of each.

3

Solve a Real Problem

30 marks

Choose an education problem and solve it step-by-step using the 5 stages.

4

Barriers & Solutions

15 marks

Name at least 1 barrier that stops the solution and how to overcome it.

+ Presentation (10 marks) — Be clear, confident and visible on camera

PART 1 — PROBLEM-SOLVING TOOL: THE FLOWCHART

25 Marks

WHAT IS A FLOWCHART?

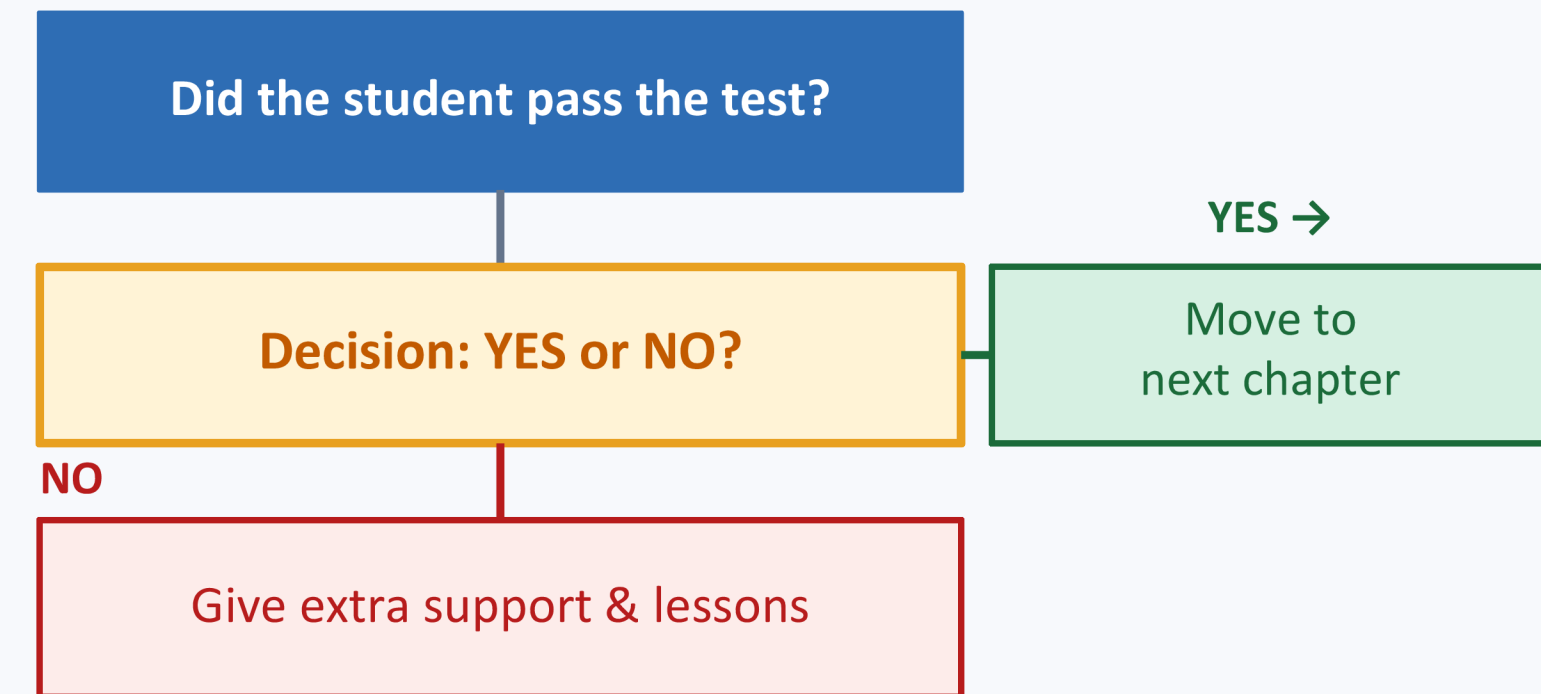
A flowchart is a diagram that shows steps in a process using shapes and arrows.

Rectangle = an action or step

Diamond = a YES/NO decision

Arrow = which direction to go next

It helps you see the whole process at once so you never miss a step.



Example Flowchart: Teacher decides if a student needs extra help

SAY THIS IN YOUR VIDEO:

"A flowchart is a diagram that uses shapes and arrows to show steps in order. It helps us make decisions clearly at every stage. For example, a teacher can use a flowchart to decide whether a student needs extra support — starting with the question: did the student pass the test?"

PART 2 — THE 5 STAGES OF PROBLEM SOLVING

20 Marks

1

Stage 1 — IDENTIFY THE PROBLEM

Recognise something is wrong. Name it clearly.

Example: "30% of students are failing their tests."

2

Stage 2 — UNDERSTAND THE PROBLEM

Ask WHY it is happening. Find the root cause.

"Students say lessons move too fast and are hard to follow."

3

Stage 3 — THINK OF POSSIBLE SOLUTIONS

Brainstorm different ways to fix the problem.

"Slow down lessons / Use visual aids / Introduce peer tutoring."

4

Stage 4 — CHOOSE & APPLY BEST SOLUTION

Pick the most practical idea and put it into action.

"Use visual aids AND peer tutoring twice a week."

5

Stage 5 — EVALUATE — DID IT WORK?

Check if the problem improved. If not, try a different solution.

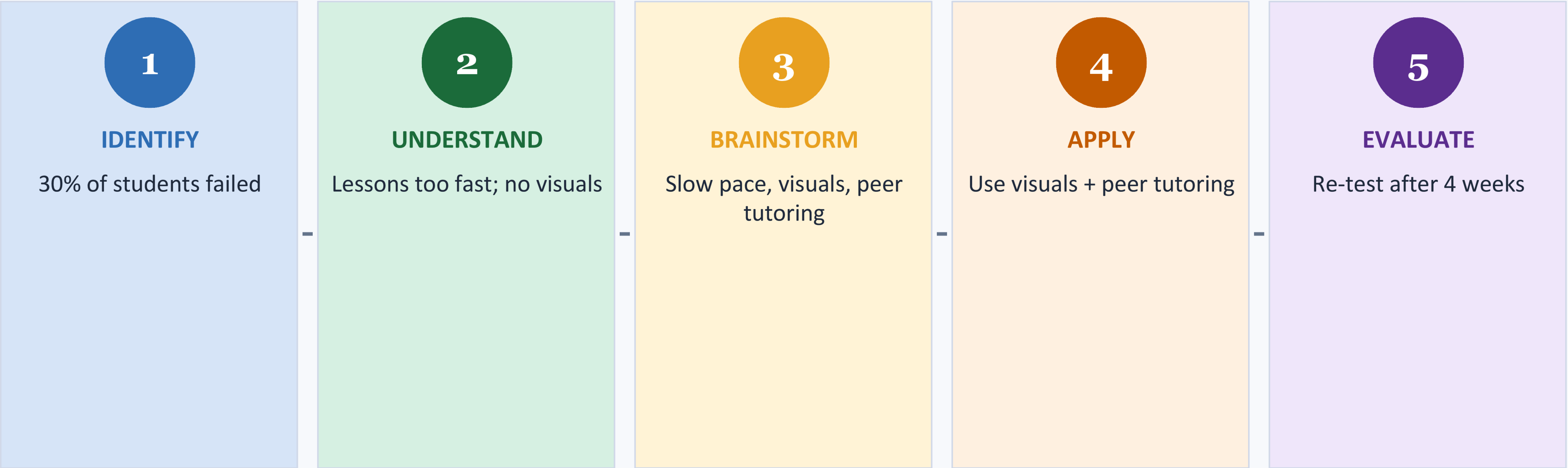
"After 4 weeks, 70% of students now pass. The solution worked!"

PART 3 — SOLVING A REAL EDUCATION PROBLEM

30 Marks

THE PROBLEM

Many students in a school are failing their tests because they do not understand the work being taught in class. 30% of Grade 10 students failed the last assessment.



PART 3 — STEP-BY-STEP WALKTHROUGH

1

Stage 1 — IDENTIFY

"30% of Grade 10 students failed their last test. This is higher than normal."

2

Stage 2 — UNDERSTAND

"After speaking to students, I found lessons move too fast and there are no visual aids."

3

Stage 3 — BRAINSTORM

"Option A: Slow down. Option B: Use visual aids. Option C: Introduce peer tutoring."

4

Stage 4 — APPLY

"I combined visual aids AND peer tutoring sessions twice a week."

5

Stage 5 — EVALUATE

"After 4 weeks, 70% passed the re-test. The failure rate dropped significantly."

PART 4 — BARRIERS AND HOW TO OVERCOME THEM

15 Marks

A **BARRIER** is anything that **STOPS** your solution from working properly.

1

BARRIER 1: Lack of Resources

✗ THE BARRIER:

The school does not have enough projectors or printed materials for visual aids.

✓ HOW TO OVERCOME IT:

Use free digital tools like Google Slides on a single screen. Ask students to draw diagrams in their notebooks.

2

BARRIER 2: Resistance to Change

✗ THE BARRIER:

Some teachers don't want to try peer tutoring or new visual methods — they prefer their old way.

✓ HOW TO OVERCOME IT:

Hold a short staff workshop showing the improved test results as evidence. When teachers see data, they are more likely to accept change.



CHECKLIST — BEFORE YOU RECORD



I know what the 4 parts of my video are



I can name a problem-solving tool and explain it with an example



I can explain all 5 stages of problem solving



I have chosen an education problem (e.g. students failing)



I can walk through the problem step-by-step using the 5 stages



I can name at least 1 barrier and how to overcome it



I have a flowchart ready to show on camera



I have practised my video at least once before recording



My video will be 5–7 minutes and saved as MP4

If you can tick all boxes, you are ready to record!



MARKS BREAKDOWN

Part 1

Problem-Solving Tool



25 marks

Part 2

Stages of Problem Solving



20 marks

Part 3

Real Education Problem



30 marks

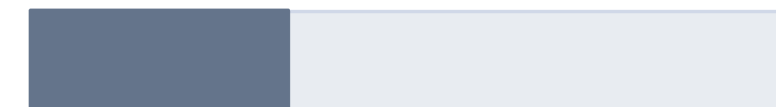
Part 4

Barriers & Solutions



15 marks

Presentation & Clarity



10 marks

TOTAL

100 marks



YOU ARE READY!

Follow the 4 parts. Use the script. Speak clearly.
You have everything you need to succeed.

 Tool

 5 Stages

 Real Problem

 Barriers

 Record!